Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering
09/2023 – Expected 06/2028

- GPA: 2.93/4.00
- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi, Regional Organizational Team Member at ESTIEM; participated in Work and Travel USA; Student-Athlete (sailing, skiing & windsurfing), doing wakeboarding, kiteboarding, surfing, bouldering, running, fitness, and traveling as hobbies.
- Volunteer: Volunteering Division Member at Industrial Engineering Club (BÜYAK), gave educational mentorship for high school students.

Kabataş Boys' High School
09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.
- Volunteer: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX
09/2026 – Present

- Joining the backend development team to contribute to the design and implementation of scalable software solutions. Excited to take on responsibilities involving robust backend architecture, API development, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Intern at Akbank
07/2026 – 09/2026

- Completed a 2-month internship with the SRE team, contributing to the reliability, scalability, and performance of mission-critical banking systems. Took on responsibilities involving infrastructure operations and systems engineering in a large-scale enterprise environment.

Software Engineering Intern at Smash Mate
06/2026 – 07/2026

- Completed an intensive 1-month internship during the critical pre-launch phase, contributing to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native). Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Played an active role in agile workflows, code reviews, and system optimizations, leading directly to the product's successful official launch.

Technical Skills & Projects

- **AlphaTrigger** (Python, FastAPI, Next.js, WebSockets) — Autonomous trading simulator with Z-Score mean reversion & volatility sizing.
- **FinTech Platform** (Python, FastAPI, Next.js) — Real-time dashboard with limit-order matching engine & portfolio risk analytics.
- **Seated** (Next.js 15+, TypeScript, Prisma, PostgreSQL) — Multi-tenant SaaS for real-time restaurant floor plans, reservations & operations.
- **Performo** (TypeScript, Next.js, Express, Tailwind CSS) — Administrative console with role-based access control (RBAC) & task workflows.
- **Enterprise FAQ** (Next.js, TypeScript, Express, SWR) — Enterprise knowledge base with optimistic UI updates, caching & error boundaries.
- **Space Shooter IPC** (C11, POSIX FIFOs, Signals, dlopen) — Asynchronous 60 FPS input engine using non-blocking FIFOs and signals.
- **SchedSim** (x86-64 GNU Assembly, Linux Syscalls) — Low-level CPU scheduling simulator executing FCFS, SJF, SRTF, and Round Robin.
- **Solar Escape** (RISC-V RV32I Assembly, Spike) — Bare-metal pathfinding algorithm with greedy route selection & manual stack framing.
- **CoreMark Profiling** (RISC-V, CoreMark) — Report benchmarking compiler optimizations (-O0 to -Ofast) and ISA pipeline throughput.
- **Spireslayer** (C, POSIX, Makefile) — Unix command interpreter featuring a custom backtracking recursive descent parser.
- **Aura** (Python) — Functional programming language interpreter supporting closures, lexical scoping, and currying.
- **MatrixNet** (Java, Graph Theory) — Network topology analysis tool implementing Tarjan's SCC algorithm & articulation points.
- **GigMatch-Pro** (Java, Priority Queues) — Algorithmic job-matching engine leveraging max-heaps for priority-based candidate allocation.
- **Smash Minesweeper** (React Native, TypeScript, Expo) — Mobile puzzle game built with reducer-driven state engine and BFS flood-fill.
- **Qt 2048** (C++, Qt6, WebAssembly) — Cross-platform game compiled to WebAssembly for browser deployment with state undo.
- **Nightpass** (Java) — Terminal-based strategy survival card game driven by a deterministic finite state machine.

Tools & Languages

- **Tools & Software:** Git/GitHub, AI-Assisted Development, MS Office
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)